

Dark Forest (Dense Forest / Night) Map Design

Executive Summary

This Concept is designed as a balanced and tactical map for Thermal warfare, sensor detection, stealth movement, and information denial. The terrain features thick canopy cover, scattered undergrowth, mossy cliffs, abandoned research cabins, shallow creeks, and elevated ridges. Darkness, humidity, and foliage create extreme visibility challenges. The map emphasizes Obsidian Protocol's autonomous light combat behavior, Thermal sensor fade, covert-recon warfare, and tactics under low-light constraints. In AR (Real Life) the map anchors at 4-5 m in-shade distance, covering a 10x10 m physical area representing ~1 km² of hypothetical forest terrain. Key features include Thermal chokepoints, sensor blind zones, hidden trails, and ambush-heavy infiltration routes.

Design Goals & Gameplay

Night Combat & Thermal Warfare

Dark Forest provides aggressive sensor thermal imaging IR sensors, and low-light optics deniable. Units rely on heat signatures rather than visual clarity.

Autonomy & Low-Visibility Pathfinding

Autonomy & sensor concealment, avoid mossy coverings, and use darkness for stealth. Units adapt for drifting fog density and thermal interference.

Sensor & Information Warfare

Sensor creep, thermal fog, and temperature gradients distort sensors. Relay towers on ridges reduce clarity. Drifting fog density creates massive thermal blind zones.

Logistics & Resources

Relay towers, research equipment, and forest deposits serve as supply nodes. Supply convoys must navigate narrow trails and avoid ambush-prone dense pockets.

Deployment & Progress

A heat signature trigger initiates a fog density increase when sensors are active. Fog density increases between stacks of entry, thermal drones, or light vehicles. Blue displays from the southwest night-up camp. Red displays from the northeast ridge bunker.

Tracking & Occlusion

Heat signature detection is limited to sensor planes. Depth-mesh (if available) provides limited occlusion. Visual objects render over real-world scene, assume minimal real-world occlusion.

Spatial Constraints (Nreal AR)

Device Capabilities

Screen: 1080p x 1080p @ 90Hz, 1080x1080 per eye resolution. Optical viewing distance ~4-5 m.

Playable Area & Scaling

Playable area: 10m x 10m (100m²) forest zone (1:100 scale). Designed for tabletop AR.

Layout Overview

Coordinate System

000-1000 grid (1 unit = 1 m). Origin (0,0) = southwest corner.

Key Locations

Blue HQ (North-East) — 0,000 - 0,000 (0,0) — Thermal sensor blind zone with mossy forests.

Red AR (West-South) — 1,000 - 1,000 - 1,000 (0,0) — Thermal sensor blind zone with mossy forests.

Central Research Cabin — 0,500 - 0,500 - 0,500 (0,0) — Thermal sensor blind zone with mossy forests.

Research Outpost — 0,500 - 0,500 - 0,500 (0,0) — Thermal sensor blind zone with mossy forests.

Old Forest Cabin — 0,500 - 0,500 - 0,500 (0,0) — Thermal sensor blind zone with mossy forests.

Creek Network — 0,500 - 0,500 - 0,500 (0,0) — Thermal sensor blind zone with mossy forests.

Hidden West Trail — 0,500 - 0,500 - 0,500 (0,0) — Thermal sensor blind zone with mossy forests.

Thermal Ridge A — 0,500 - 0,500 - 0,500 (0,0) — Thermal sensor blind zone with mossy forests.

Thermal Ridge B — 0,500 - 0,500 - 0,500 (0,0) — Thermal sensor blind zone with mossy forests.

Neutral Forest Outpost — 0,500 - 0,500 - 0,500 (0,0) — Thermal sensor blind zone with mossy forests.

Chokepoints & Sightlines

Hidden Night Trail

Perfect for ambushes, extremely narrow and concealed.

Creek Coverage

Water creek flow (ambushes, thermal sensors denied).

Cabin Cluster

Interior building routes, close-range engagements.

Shadow Basin

Heavy foliage + darkness + unpredictable sightlines.

Sightlines

Thermal Ridge A provides long-range sensor coverage (7-100 m). Fog and temperature gradients create thermal blind pockets.

Deployment Zones

Blue Deployment: 100m² area (100m x 100m) — 100m x 100m.

Red Deployment: 100m² area (100m x 100m) — 100m x 100m.

Neutral Forest Zone: Outpost at 0,500 - 0,500 - 0,500 (0,0) for dense forest.

Assets & Environment

Terrain

Forest floor, undergrowth, creeks, ridges, fallen logs.

Vegetation

Deciduous trees, shrubs, ferns, moss, fog pockets.

Buildings & Props

Military: IR floodlights, antennas, bunkers. Civilian: Cabins, sheds, research equipment. Logistics: Trucks, crates, fuel drums, thermal drones.

Effects

Fog, humidity haze, drifting mist, falling leaves, thermal distortion.

Audio

Wind, insects, distant machinery, radio chatter, night wildlife.

LOD & Budget

Texture Budget

2048 for terrain/buildings, 1024 for props. DXT5/LC compression. Target 4000 MB.

Unreal / Nreal Integration

Engine & Plugins

Unreal Engine 5 + Nreal SDK (ARKit/ARCore enabled).

Rendering

Forward renderer, stationary lights, baked GI, aggressive culling.

Anchor

Single spatial anchor for entire map.

Interaction

Phone pointer for playback, HUD as screen-space widgets.

Multiplayer

Client-server architecture, offload AI/physics to server.

Implementation Plan & Testing

Milestones

1. Prototype Level — Month 1: Block-out forest terrain, ridges, cabins, AR anchor test.

2. Core Features — Month 2: ARKit/ARCore, thermal sensor configuration, autonomous behaviors.

3. AR Interaction — Month 3: Anchor, vision distortion, basic interactivity.

4. Content Fill — Month 4: Final models, textures, lighting, audio.

5. Optimization & Build — Month 5: Performance, balance, refine thermal detection.

6. Testing & Ship — Month 6: Final build, release test, sensor calibration.

Testing Plan

Autonomy

Verify stealth logic, thermal sensor navigation, ambush behavior.

Comms/Sensors

Test relay detection, fog-of-war updates, LOS checks under night conditions.

Performance

Address render on target device.

QA Checklist

Temperature, camouflage behavior, thermal response, deployment budget enforcement, anchor drift, sensor accuracy.